

Andi Miller - Big Data Systems Engineer

Email	Website	Citizenships	Portfolio
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Skills and Experience

Languages

Scala	13 years of experience, with 10 years of purely functional development, current main language.
Java	Experience with backend Java; Apache Pinot contributions
C	Commercial experience writing patches for OpenSIPS and Asterisk while at Gradwell.
Python	Used for many projects including open source infrastructure for EVE Online alliances.
Haskell	Personal projects and small work tooling.

Technologies

Functional Programming	Experience writing purely functional software in multiple languages.
Stream Processing	Kafka, zeromq, RabbitMQ, Pulsar and other high throughput streaming technologies.
Big Data	Druid, Pinot, Elastic, Cassandra and Spark for storage and aggregation.
Data Sketches	HyperLogLog plus variants, Theta, Tuple, etc.
Artificial Intelligence	Classification and Regression model training, building with LLMs, MCP, etc.
Distributed Systems	Have built concurrent distributed systems with actors and consensus systems.
Language Development	Experience implementing domain specific languages with technologies including ANTLR, fastparse and droste.
Observability	Metrics, tracing, alerting, etc.

Interesting Work

Patent US10579827B2

- Optimisation for HyperLogLog which reduces memory use by sharding around the cluster.
- Included lazy bucketing and pre-allocation of the sharded data.

IDML

- Open source JSON processing language used for data processing at DataSift and Meltwater
- JVM first, integrated with Kafka Streams, Hadoop and Elasticsearch.
- Implemented with ANTLR and Scala.

Open Source Contributions

http4s

- added OkHttp client
- Header class QoL improvements

apache/pinot

- expanded Theta Sketch support
- expanded HyperLogLog support
- added Tuple Sketch support
- added UltraLogLog support

typelevel/cats

- added extra syntax for Bitraverse

typelevel/cats-effect

- added mapK to Resource

Employment

Senior Software Engineer, Permutive, 2022 - Present

- Scala / Typelevel / Agile
- Extended insights products, rebuilt on Pinot, average query time 30s => 1s, costs 4x lower.
- Rebuilt data query layers around Pinot and Iceberg.
- Built out MCP product for customers.
- Added LLM embedding-based advertising cohorts to contextual product.

Senior Software Engineer, Meltwater UK, 2018 - 2022

- Scala / Typelevel / Agile
- Practiced DevOps maintaining our own infrastructure and being on call.
- Practiced Type Driven Development and Test Driven Development.
- Maintained an in-house Domain Specific Language for transforming JSON at scale in Scala.
- Worked on ingestion of firehoses into Meltwater's systems.
- Helped improve and develop the Meltwater Public API

Platform Engineer, DataSift, 2013 - 2018

- Developed microservices in Scala with a combination of Dropwizard and scalaz-based stacks, switching to typelevel later.
- Worked in agile teams using scrum development practices.
- Co-developed an in-house Domain Specific Language for transforming JSON at scale in Scala.
- Worked on privacy-first analysis of Facebook data in Facebook's datacentre.
- Worked on the DataSift open source client libraries, with a focus on the Python one

VoIP Engineer, Gradwell dot com Limited, Placement Year, 2011 - 2012

- Maintained and developed a large award-winning Open Source VoIP stack
- Regularly worked with both Support and Sysadmin departments
- Worked in a variety of languages, including Perl, C, Python and Ruby
- Helped migrate from CVS to Mercurial
- Helped move to and administer Jira
- Re-engineered the VoIP platform to work across multiple datacentres.
- Developed deployment infrastructure with puppet to reduce deployment time from 10 hours to 1.

Freelance Coder (Google Summer of Code), Google, 2010

- Did some experimental work on integrating open source video and audio editors Pitivi and Jokosher
- Worked with GTK and DBus

Education

B. Eng. Software Engineering, University of Wales Aberystwyth, 2009 - 2013

- 2:1
- Averaged over 70% in Programming modules.
- Opted into Open Source and Artificial Intelligence modules.